EVALUATING ARSENAL DEPTH DIVERSITY USING MOVEMENT ENTROPY

By

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A Personal Project That Does Not Serve as a Form of Work For Any Organization

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ABSTRACT

Current methods that revolve around evaluating what a pitcher has in their repertoire typically revolve around the number of different pitch types, based on Statcast classifications, they throw. These methods tend to not consider true movement diversity that a pitcher displays. **PURPOSE:** The purpose of this analysis was to look at whether entropy within a pitch arsenal can be used as an effective predictor for success. **METHODS:** 291 MLB pitchers (Arsenal Depth: 5.27 ± 1.34 pitch types, Entropy: 6.81 ± 0.34) were evaluated in quantity of pitches within their arsenal and entropy of their pitch movement using Kernel Density Estimation (KDE). This data was then used in a linear regression against a variety of performance metrics that are used to evaluate a pitcher's effectiveness. **RESULTS:** Entropy alone held no significant relationships with performance statistics that were considered except for xSLG, although strength of relationship was very weak as were the rest of the relationships. Arsenal depth alone was strongly correlated with all performance metrics and held moderately strong relationships with xSLG, K%, and xWOBA. The direction of all the relationships between arsenal depth and performance statistics suggested a decrease in pitching performance as arsenal depth increases. Entropy/arsenal held very similar strengths as arsenal depth alone, however with opposite directions, suggesting an increase in pitching performance as entropy/arsenal increases. **CONCLUSION:** While the results were purely correlational and not causal, they suggest the potential for a new method in considering how a pitcher's arsenal is evaluated, as it seems that using entropy/arsenal may be an effective manner for doing so.

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INTRODUCTION

It is no secret that Major League Baseball (MLB) is subject to trends, especially with the growth of research and development to become such an integral part of player development, and trends within pitchers tend to be the driving factor with their role as an instigator in the game itself. One of the more recent trends that gained notoriety in the 2025 MLB season was the growth of the average pitcher's arsenal. While baseball has since grown from the traditional 3 pitch mix of a fastball, change-up, and breaking ball variant, the 2025 season shined a spotlight on just how large and diverse these pitch mixes can get. Most notable with the amount of success seen by Paul Skenes displaying 7 pitches (6 of which at a usage of at least 5% and considered to perform at an above average to elite level in terms of run value; *Paul Skenes Stats: Statcast, Visuals & Advanced Metrics*). Despite a less successful season than the previously mentioned Skenes, Yu Darvish is likely seen as one of the true originators of the large pitch mix, as he stated in a 2021 clip from an interview with Rob Friedman, more commonly @PitchingNinja on social media, that he throws 11 different pitches (*Rob Friedman on Instagram: "Yu Darvish throws 11+ unique pitches* 😯*"* 2021). While Statcast doesn't recognize all 11 pitches he claims to have in his arsenal, as some pitches he claims to throw are the same pitch but at different speeds, Statcast does recognize an impressive 9 different shapes (7 of which had a usage of at least 10%) that he had shown in that 2021 season (*Yu Darvish stats: Statcast, Visuals & Advanced Metrics*). In his 2025 article titled, "Is 2025 really the 'year of the pitch mix'?", David Adler, Senior Coordinator of Data & Research at MLB, outlines the trend that MLB has seen in pitch arsenal sizes in the Statcast era, displaying that in 2025 a record-high 253 starting pitchers were throwing at least 5 different pitch types as classified by Statcast and, even further, 125 starting pitchers were throwing at least 6 different pitch types, also a record-high, illustrated in Figure 1.

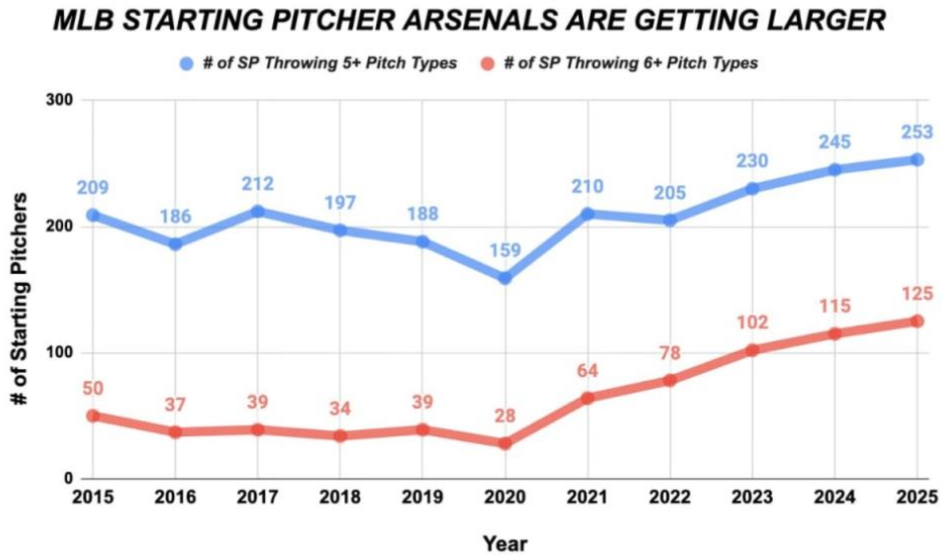

Figure 1 – *Growth in the number of starting pitchers throwing at least 5 and at least 6 pitches from 2015-2025 (Reprinted from Adler, 2025)*

Adler further notes that this trend in arsenal expansion is apparent not only in starting pitchers, but relief pitchers as well. He displays this with his use of Tom Tango's "pitch palette" metric, that factors in the usage of each pitch in a pitcher's arsenal to quantify their arsenal size, interpreted as larger numbers indicate greater arsenal sizes, while lower numbers indicate a smaller arsenal. Consistent with the results in Figure 1, starting pitchers showed a record-high pitch palette in 2025 while relief pitchers did the same, as shown in Figure 2.

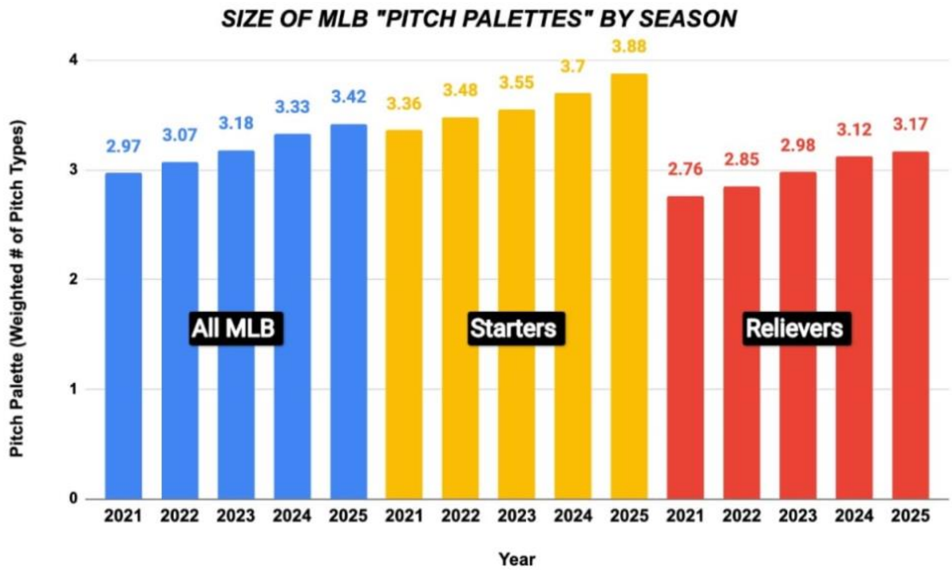

Figure 2 *- Growth of MLB pitchers' "pitch palettes" from 2021-2025 (Reprinted from Adler, 2025)*

Problem Statement

While this growth in arsenal size is apparent on a year-to-year basis as previously outlined, current measures only look at the number of different pitch types in a pitcher's repertoire based on classifications along with their usage of each pitch. Although classifications separate different pitches based on their movement profile, the classifications inherently can't fully capture the diversity of pitches from one another due to their nature of containing a range of accepted movement profiles. Two pitchers may each throw five pitches, but one of them may have 5 different pitches that are very distinct from one another, while the other may throw 5 different pitches that have fairly similar movement profiles. In situations such as this, arsenal size sees two pitchers that are identical to one another, failing to consider a very important distinction between the two. This is where the use of entropy, as a tool for evaluation, can come into play. Entropy is derived from information theory and describes "…a measure of the average uncertainty in the random variable" (Cover & Thomas, 2006, p. 5). In the context of pitch movement, entropy can describe how broadly pitch movement is distributed throughout its two-dimensional movement plane.

Purpose

The purpose of this analysis was to look at whether entropy within a pitch arsenal can be used as an effective predictor for success.

Research Questions

This analysis stemmed from one original research question that is outlined in the subheading that follows. While considering the results from this question, this analysis branched into a total of three separate research questions that all built off results from the question that preceded it.

Research Question #1: Can entropy effectively be used to describe a pitcher's arsenal and predict performance?

In response to this question, the results, which will be further described in the "Results" section, provided little significance, leading to the idea that entropy may have too much noise to find a meaningful relationship.

Research Question #2: With entropy showing minimal relationship with performance, is arsenal depth alone a better predictor for performance?

These results, again to be further discussed in the "Results" section, provided results that were, while more meaningful, inconsistent with the intuition that comes with the trend in increasing pitch arsenal depth. This led to wondering whether some form of entropy as a function of arsenal depth could better function as a performance predictor.

Research Question #3: Can entropy as a function of arsenal depth serve as a predictor of performance?

METHODS

This section will outline the methodology used to answer the aforementioned research questions. As previously mentioned, the purpose of this analysis was to look at whether entropy within a pitcher's arsenal can be used as an effective predictor for success.

Data

Inclusion criteria for pitchers in this analysis was solely whether they recorded a minimum of 1000 pitches thrown during the 2025 season. This yielded a sample of 291 pitchers to be considered for analysis.

Pitch Movement Data

All pitch data was recorded from MLB's Statcast in-game tracking technology and was retrieved using the pybaseball (version 2.2.7) package in Python (version 3.11.13).

Performance Data

All performance data was retrieved from Baseball Savant using their Custom Leaderboard feature (*Statcast custom leaderboards*). With FIP not being available on Baseball Savant's list of statistics to include, it was calculated within Python using the formula provided by FanGraphs *(Slowinski, FIP)* and the FIP constant of 3.135 for the 2025 season was obtained from FanGraphs' "Guts!" table (*Guts!*).

Statistical Analyses

All tables, analyses, and plots were done in Python using the following packages and (version): matplotlib (3.11.1), numpy (2.5.1), pandas (3.0.5), plotly (6.9.0), scikit_learn (1.9.0), scipy (1.18.0).

Entropy

Using a random sample of 1000 pitches from each pitcher during the 2025 season, pitch movement density was estimated across the two-dimensional pitch movement space (horizontal and induced vertical break). A sample size of 1000 pitches was chosen to ensure comparable entropy estimates while minimizing sample variability. Since pitch movement exists continuously within this two-dimensional space, a continuous estimate of pitch density was required before entropy could be calculated. This is where a Gaussian kernel density estimator (KDE) was deemed to be useful via SciPy's gaussian_kde() function. The resulting density surface was normalized so that each grid cell represented the probability of observing pitch movement at that location. Shannon entropy was then calculated as $-\sum_i p_i \log(p_i)$, where p_i represents the normalized density at each grid cell. Using Shannon entropy allowed the individual density values across the whole movement space to produce a single value that can describe the diversity of a pitcher's arsenal. This metric can be interpreted as a measure of how diverse a pitcher's movement profile is, with higher values reflecting a broader distribution of movement profiles and lower values reflecting movement concentrated in fewer regions. Importantly, entropy is influenced not only by the number of distinct pitch clusters and how frequently each cluster appears, reflecting the overall uncertainty of pitch movement rather than simply the number of pitch types.

Arsenal Depth

Arsenal depth (referred to in the data as 'arsenal_count') was determined by counting the number of unique pitch types that each pitcher displayed in their Statcast data, dependent upon how Statcast classifies different pitch types. This metric was meant to replicate the trend seen in arsenal depth in the MLB, with a higher value simply meaning that the pitcher throws more pitch types and, in theory, providing more different movement offerings for the hitter to have to account for.

Entropy/Arsenal

Entropy/Arsenal (referred to in the data as 'entropy/arsenal_count') was calculated by simply taking each pitcher's entropy value and dividing it by their arsenal depth (arsenal_count). Entropy/arsenal is intended to display the variability in movement for every movement offering each pitcher has, a large value can be interpreted as a large amount of diversity, or spread, between pitch classifications.

Performance Statistics

The statistics chosen to evaluate each pitcher's performance, or "success", were strikeout percentage (K%), walk percentage (BB%), expected weighted on-base average (xwOBA), weighted on-base average (wOBA), expected slugging percentage (xSLG), and fielder-independent pitching (FIP). These particular stats were chosen to represent the pitcher in a well-rounded manner in their ability to prevent offensively favorable contact (xSLG, HR/9, xwOBA), limit baserunners (K%, BB%, and wOBA), and most importantly run prevention (FIP). Stats such as earned run average (ERA), slugging percentage (SLG), and batting average (BA) were omitted or looked over for their expected alternative (xSLG instead of SLG) due to the potential for events outside of a pitcher's control to inflate or deflate that statistic.

Figure 3 *– Schematic Represenatation of Methodology*

Statcast Pitch Data

Import data, filter out pitchers with < 1000 pitches, take 1000 random samples of movement coordinates

Gaussian KDE

Calculate density surface with probabilities existing in each grid cell

Shannon Entropy

Plug probabilities into the Shannon entropy equation to find a single entropy value that represents the pitcher's arsenal

Entropy/Arsenal

Divide entropy by arsenal size

Correlation and Regression

Run correlations and regressions for entropy, arsenal size, and entropy/arsenal against each performance statistic

RESULTS

The following plots and tables provide a visual to the analyses conducted in this paper. Analysis consisted of Pearson's correlations for entropy, arsenal, and entropy/arsenal with the outlined performance statistics. Figures displaying the scatterplot relationship for each metric with its corresponding performance stats can be found in Appendix A.

Entropy

When considering entropy alone, results did not hold much significance across the board, with the exception of its relationship with xSLG. While this relationship did yield significance at the $\alpha = 0.05$ level, the strength of the relationship, consistent with the other statistics under consideration, was shown to be a very weak one. One thing to note with these relationships, however, is that the directions seem to be consistent with one another. Considering positive relationships with xSLG, xWOBA, HR/9, FIP, and wOBA and a negative relationship with K%, it seems that, while a very weak relationship, one would likely interpret it as a decrease in performance as entropy increases. The one relationship that doesn't appear to be consistent with that interpretation is the negative relationship with BB%.

Table 1 *- Pearson's r and p-values for entropy with performance statistics*

Predictor	Outcome	Pearson's r	Pearson's p-value
entropy	xSLG	0.138	0.019
	K%	-0.111	0.059
	BB%	-0.108	0.067
	xwOBA	0.107	0.069
	HR/9	0.098	0.094
	FIP	0.087	0.14
	wOBA	0.077	0.191

Arsenal Depth

Moving on from entropy, the results from arsenal count showed far more significance, as seen in Table 2, with every relationship holding significance at the $\alpha = 0.05$ level. One thing to note with these relationships is that, directionally, they match the results seen in entropy, those that suggest an increase in offensive output as the size of a pitcher's arsenal increases. This, of course, does not match the intuition of improving a pitcher's performance by increasing arsenal size that is likely behind the trend that has been outlined to be occurring in the MLB.

Table 2 *- Pearson's r and p-values for arsenal_count with performance statistics*

Predictor	Outcome	Pearson's r	Pearson's p-value
arsenal_count	xSLG	0.331	0.000
	K%	-0.305	0.000
	xwOBA	0.303	0.000
	FIP	0.253	0.000
	wOBA	0.238	0.000
	HR/9	0.214	0.000
	BB%	-0.149	0.011

Entropy As A Function of Arsenal Depth

When considering the results from entropy/arsenal, the relationships, in both magnitude and direction hold very strong significance. Similar to arsenal depth, every relationship holds significance at the $\alpha = 0.05$, however, the directions are flipped compared to arsenal depth. The directions shown as a product of entropy/arsenal seem to display a decrease in offensive production as entropy/arsenal increases.

Table 3 - Pearson's r and p-values for entropy/arsenal with performance statistics

Predictor	Outcome	Pearson's r	Pearson's p-value
entropy/arsenal	xSLG	-0.354	0.000
	K%	0.348	0.000
	xwOBA	-0.336	0.000
	FIP	-0.272	0.000
	wOBA	-0.255	0.000
	HR/9	-0.208	0.000
	BB%	0.144	0.014

DISCUSSION

The initial hypothesis for this analysis was that greater movement diversity was expected to improve pitcher performance, as the increased diversity would lead to greater uncertainty for hitters.

Entropy

As mentioned previously, entropy held no significance in its relationships with the performance statistics, except for xSLG, nor did any of its relationships show any strength. It was expected that entropy would hold a positive relationship with pitching performance, specifically, a positive relationship with K% and a negative relationship with the remaining statistics. Instead, while weak relationships, the direction was inconsistent with these expectations with the exception of BB%, as previously mentioned. This could be for a multitude of reasons, but with increases in offensive production and decreases in K%, it could be assumed that pitchers with higher entropy tend to throw their pitches in hitter-friendly zones over the heart of the plate; however, BB% does tend to be a better indicator of a pitcher's control, rather than an ability to limit offensive production. Again, these relationships are shown to be very weak and insignificant in nature so these interpretations may be premature and require further investigation. Entropy does seem to be moderately related to arsenal depth, as the relationship between the two presents a Pearson's r of 0.558 ($p < 0.001$). This was to be expected, as a pitcher that has more pitch types is intuitively going to reach more areas on their movement profile.

Arsenal Depth

Intuitively, it was assumed that the relationship between arsenal depth and performance would be no different than the expected relationship between entropy and performance, as both metrics on the surface seemed to be measuring the same thing in different ways. This was not the case, as seen in the results. Arsenal depth held a much stronger relationship across the board than entropy did, but directionally, they were identical. These relationships were not what was expected, as arsenal depth seemed to be inversely related to performance; essentially, adding more pitches caused a decline in performance. This could be accounted for by a form of survivorship bias, whereas pitchers who are struggling may be more likely to experiment with various methods of improving performance, such as introducing new pitches to increase the number of various offerings they can use. One possible explanation that could account for this relationship is the idea of pitchers being specialists, where pitchers with maybe 2 or 3 pitches that are graded as above average to elite are adding pitches that are, on paper, not as strong of offerings. This is something that Lance Brozdowski discusses in his YouTube video, titled "Pitchers Are OBSESSED With One Thing…". He, however, suggests that pitchers are doing this intentionally with a positive benefit, stating that, "…the value of adding a below-average pitch to aid in mix diversity of a pitcher's repertoire as they turn over lineups, is greater than the disadvantage of throwing a sub-optimal pitch more…" going on to discuss the idea of a pitcher's pitch mix as a whole being greater than the sum of its parts. The key idea it seems like Brozdowski is looking to portray is that this is beneficial for especially starting pitchers. In considering that, while implementing a pitch count minimum may have inadvertently excluded relief pitchers, they may still be present in the data and contributing to this relationship. Another possible explanation revolves around the concept of a pitcher's shapes blending together when they add a new pitch. It is not uncommon for a pitcher to add a new pitch that may be similar to one that already exists in their arsenal (for example, adding a cutter when they throw a primary four-seam fastball) and those pitches tend to display movement profiles that become more similar to one another, rather than two distinct pitches due to the pitcher reverting to a movement pattern they are familiar with when throwing the new pitch. They may be different enough to remain classified as two separate pitches, but the shapes are not effectively different enough for the hitter to cause discrepancy in their swing decisions. Similar to the previous discussion surrounding entropy, these proposed explanations are speculation and would require further analysis to justify their use as reasoning.

Entropy As A Function of Arsenal Depth

The creation of entropy/arsenal was in consideration of the possible explanations surrounding the results of arsenal depth's relationships with performance. Entropy/arsenal attempts to control for those two factors and combat the idea that more pitches is always better by acting as a measure for relative diversity in a pitcher's arsenal. With this, it allowed for the proposition that increasing a pitcher's arsenal size is effective only if it makes it more diverse. This idea does not completely disregard what Brozdowski proposed in promoting sub-optimal pitch shapes to increase the arsenal as a whole (Brozdowski, *Pitchers Are OBSESSED With One Thing...*) but rather acts as a check that the pitch being added truly functions as a new pitch by being diverse enough from the rest of the arsenal. Comparing the results from entropy/arsenal to arsenal depth alone, strength of relationships remained relatively constant, if not mildly improved. The key difference between the two metrics is that the direction of the relationships flipped to hold a positive relationship in pitcher performance, with increased entropy/arsenal displaying a decrease in FIP, xSLG, xwOBA, HR/9, and wOBA along with an increase in K%. There did appear to also be an increase in BB%, however, this relationship was the least strong among the other statistics (Pearson's r = 0.144).

Limitations

- entropy is totally dependent on a pitch's movement profile and does not consider other factors such as release metrics, sequencing, or command
- arsenal depth relies on Statcast's pitch classifications
    - methods for classifying different pitch types does not appear to be publicized
- analysis was limited to the 2025 season
- results are correlational, not causal

Future Work

Future analyses surrounding this topic should address the limitations listed and could include training a model to find causal relationships or that can assign weights to additional release metrics to include such as velocity, release position, and/or vertical approach angle. An analysis that extends across multiple seasons while also examining stability across those seasons could be very useful.

Average entropy within each pitch classification as a means to quantify a pitcher's ability to be repeatable for each pitch in their arsenal. An example of this would be a pitcher with an arsenal of FF, SL, CH could have a very low average entropy, inferring that there is little variation in the FF shape, SL shape, and/or the CH shape.

With mention from Brozdowski about pitch arsenal size being heavily influential for starters cycling through the order (Brozdowski, *Pitchers Are OBSESSED With One Thing...*), an interesting analysis could be done separating how starters and relief pitchers may differ in how entropy and arsenal depth is related to their performance. Along these lines, there could be work done to look at how entropy and arsenal depth is related to the average number of times a starting pitcher goes through the order in their start.

Conclusion

This analysis looked into how entropy in a pitch arsenal can be used to predict a pitcher's success. Overall, the results provided insight as to a potential new route that could be taken in evaluating a pitcher's arsenal, that being further considering an arsenal's diversity using its movement profiles, more specifically entropy. This could alter how pitch design is approached, opening the door for adding pitches based on how much they would change a pitcher's overall entropy relative to the rest of their arsenal. This project was solely exploratory rather than predictive in nature, with a goal of investigating whether movement entropy capture meaningful information about constructing a pitch arsenal and performance.

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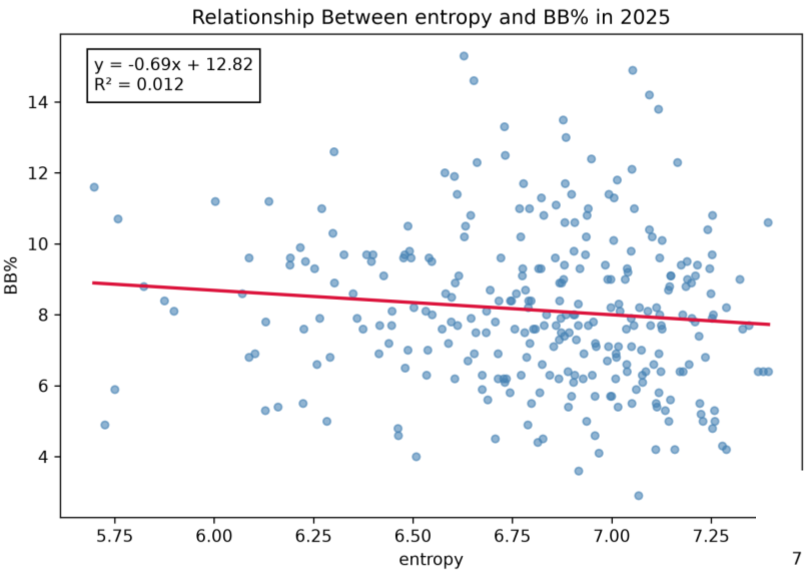
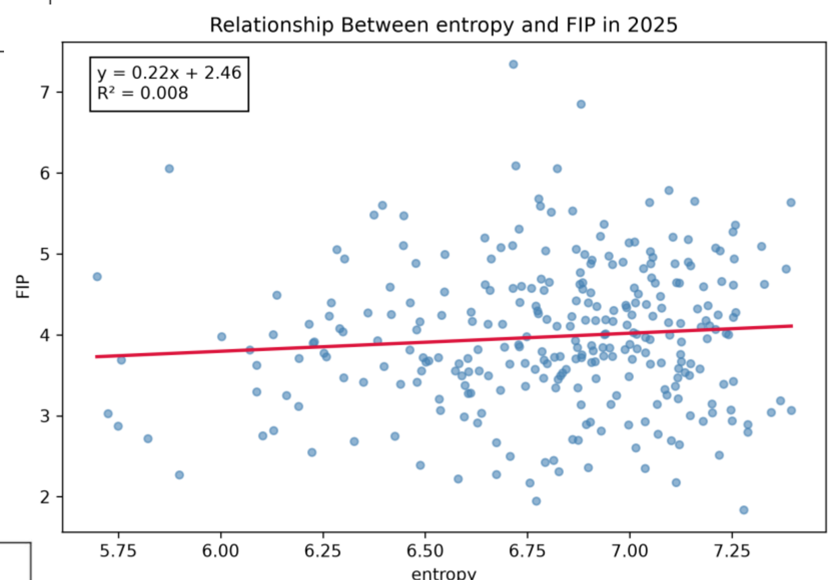
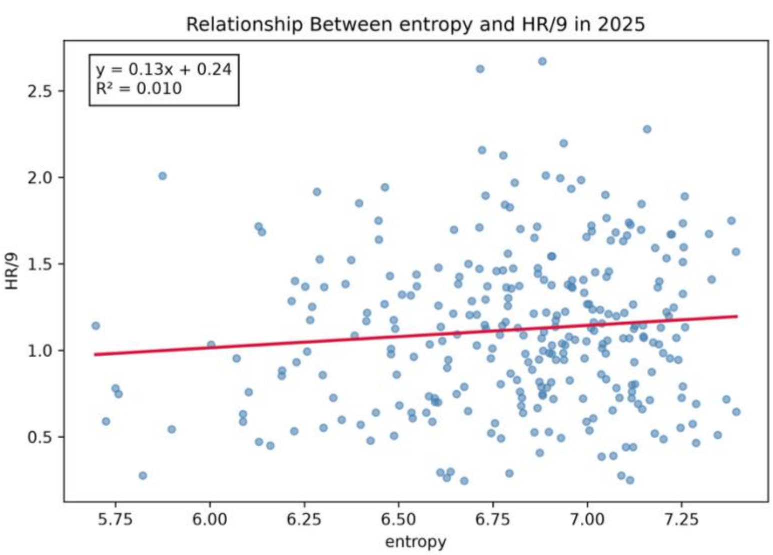
APPENDIX A

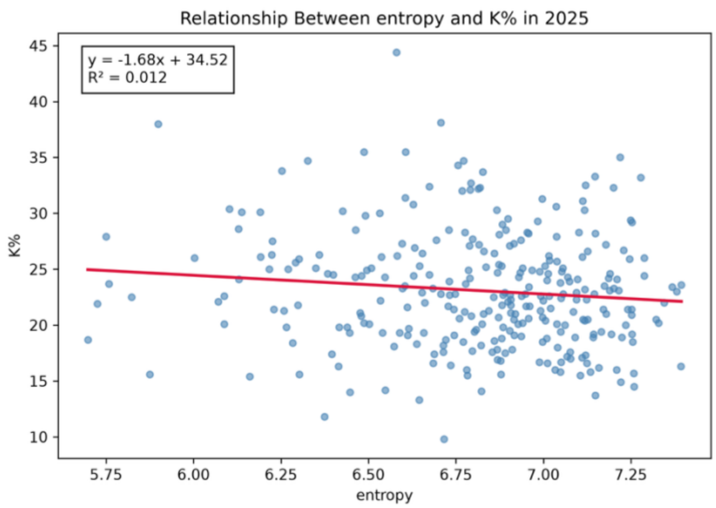
All Plots for Results

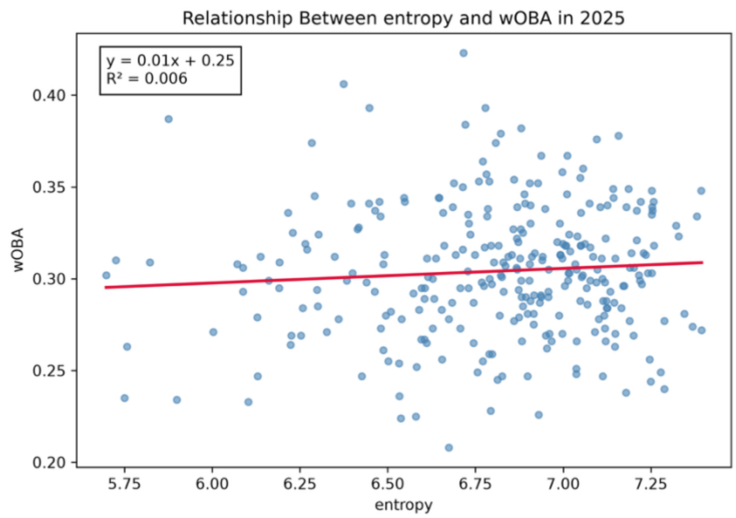
Entropy








Relationship Between entropy and K% in 2025
y = -1.68x + 34.52
R² = 0.012
K%
entropy


Relationship Between entropy and wOBA in 2025
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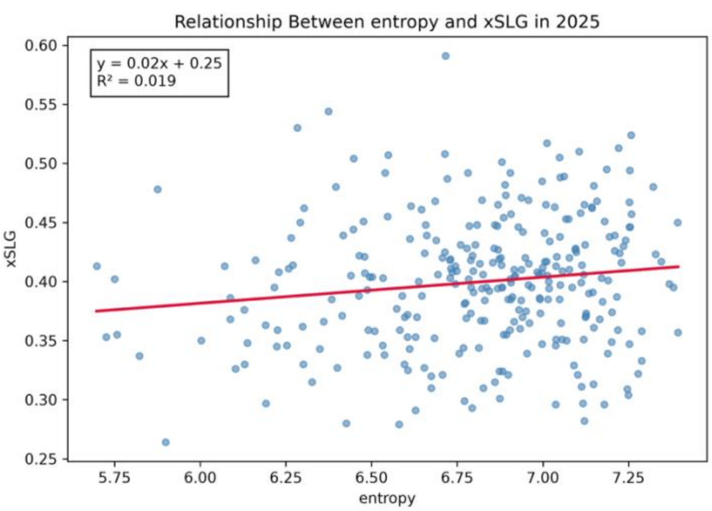
R² = 0.006
wOBA
entropy


Relationship Between entropy and xSLG in 2025
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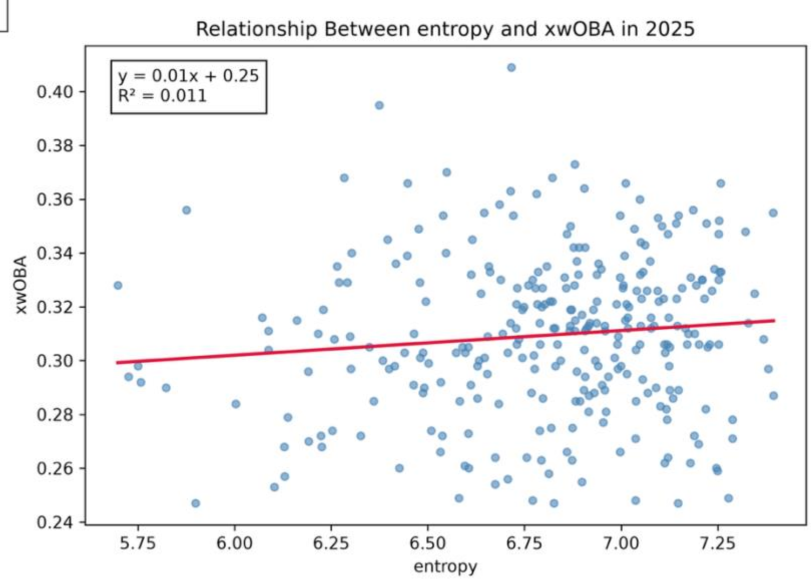
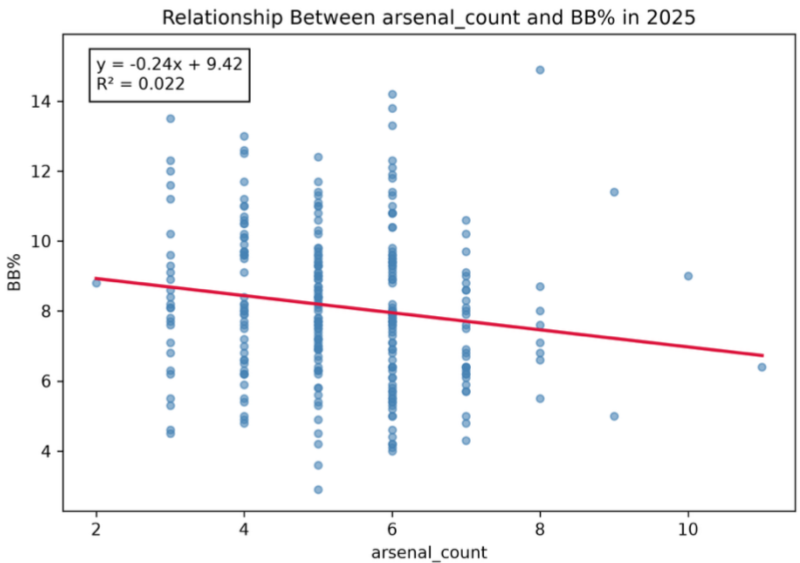
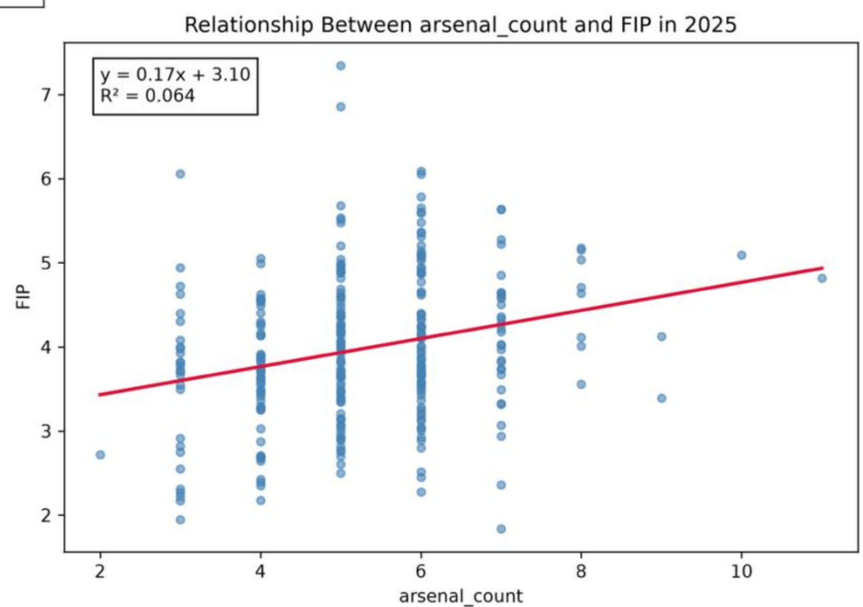
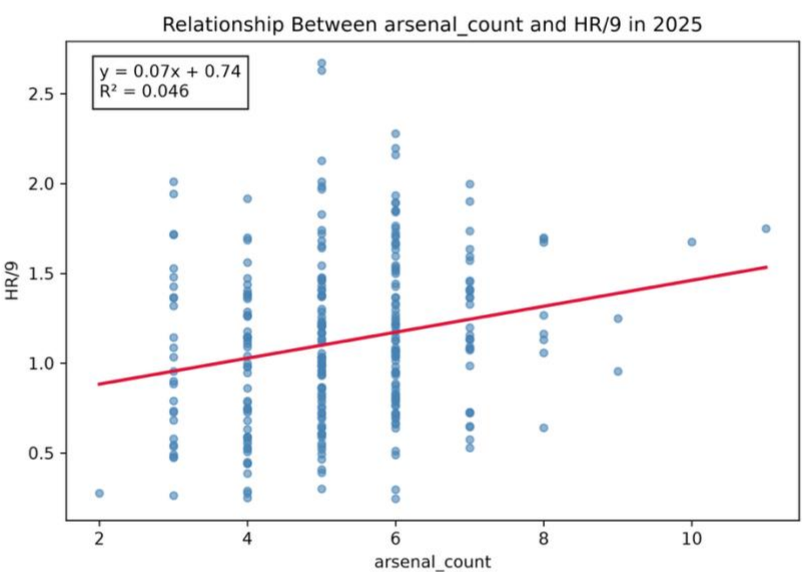
R² = 0.019
xSLG
entropy

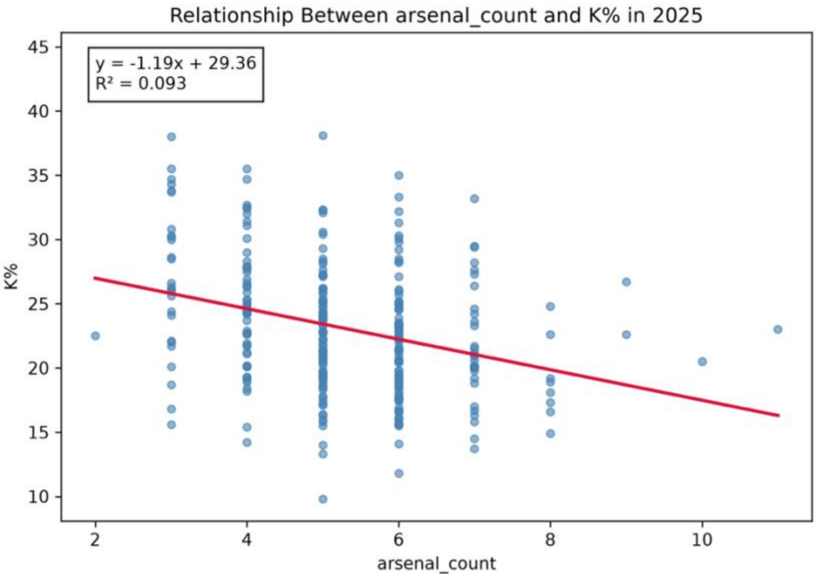

Relationship Between entropy and xwOBA in 2025
y = 0.01x + 0.25
R² = 0.011
xwOBA
entropy

Arsenal Depth








Relationship Between arsenal_count and K% in 2025
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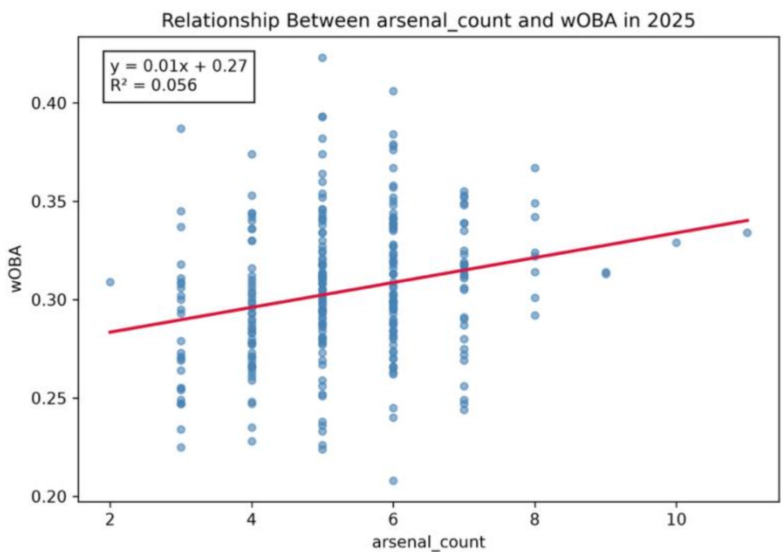
R² = 0.093
K%
arsenal_count


Relationship Between arsenal_count and wOBA in 2025
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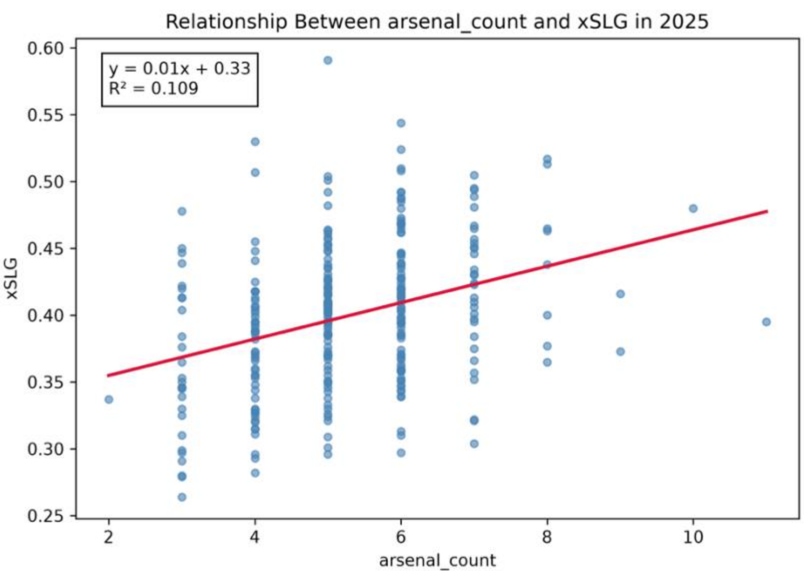
R² = 0.056
wOBA
arsenal_count


Relationship Between arsenal_count and xSLG in 2025
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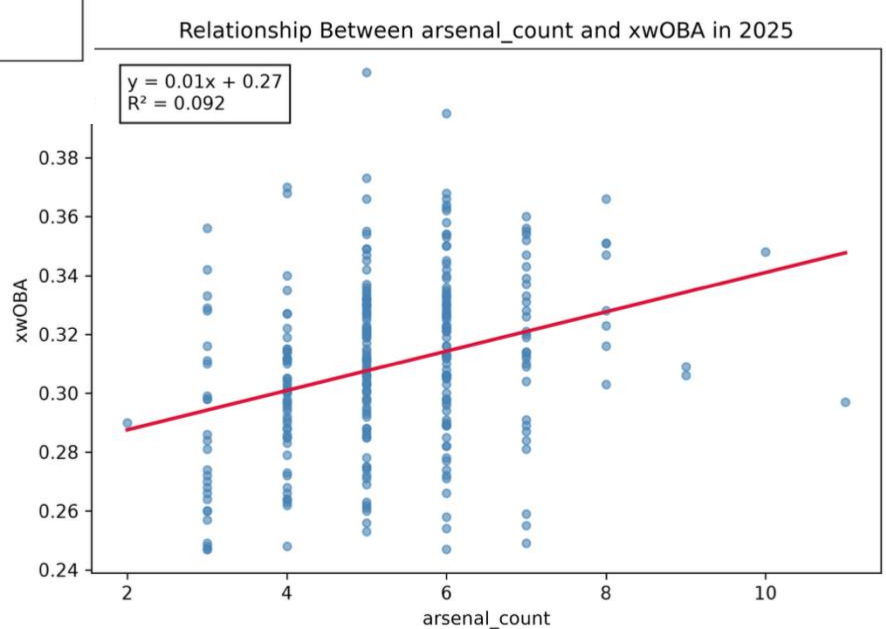
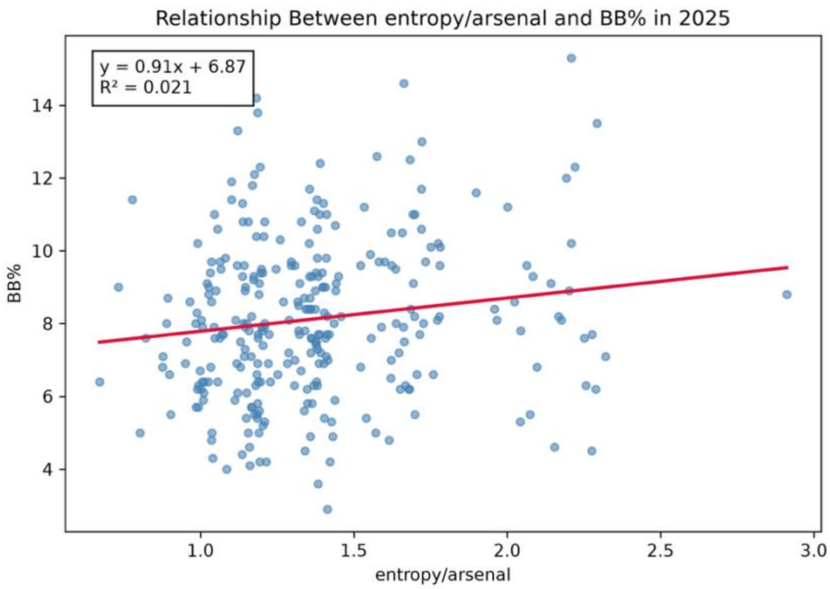
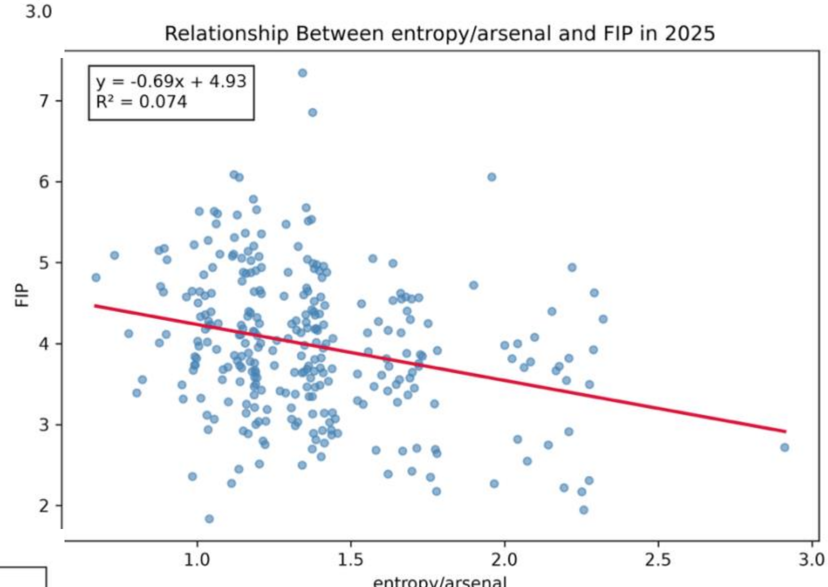
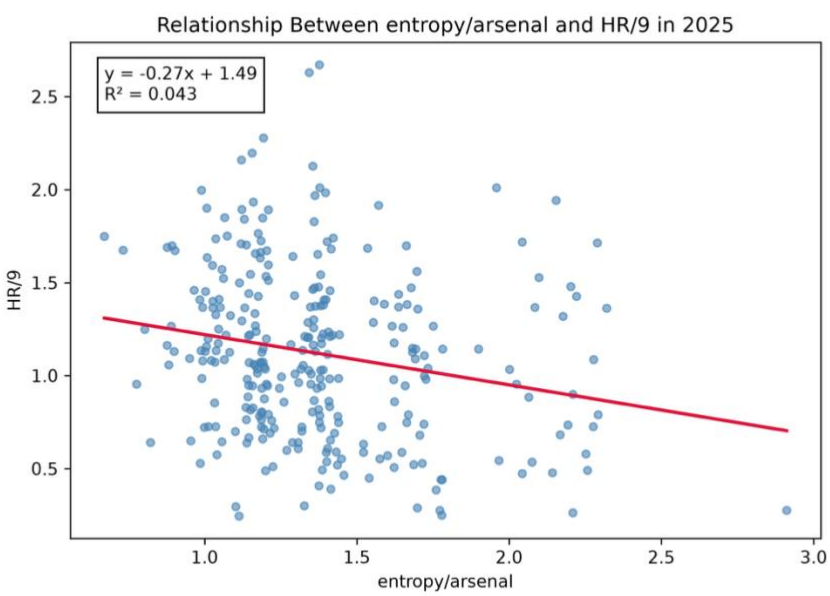
R² = 0.109
xSLG
arsenal_count

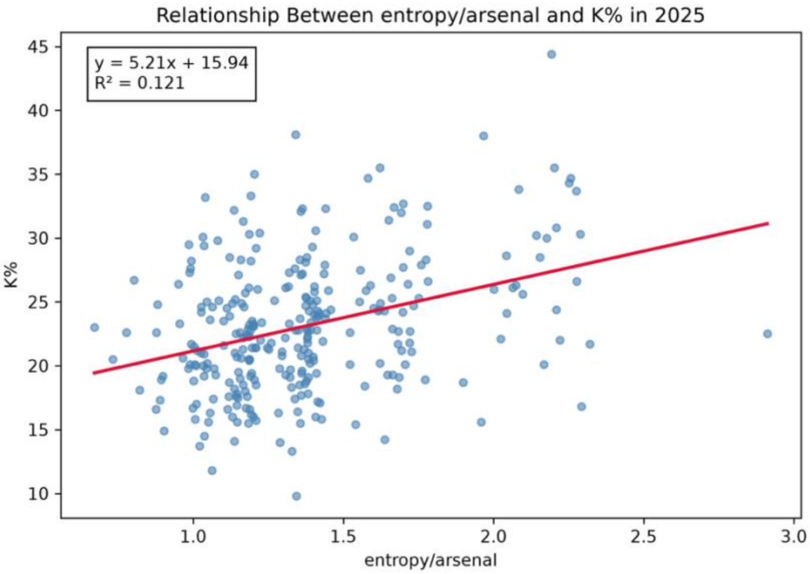

Relationship Between arsenal_count and xwOBA in 2025
y = 0.01x + 0.27
R² = 0.092
xwOBA
arsenal_count

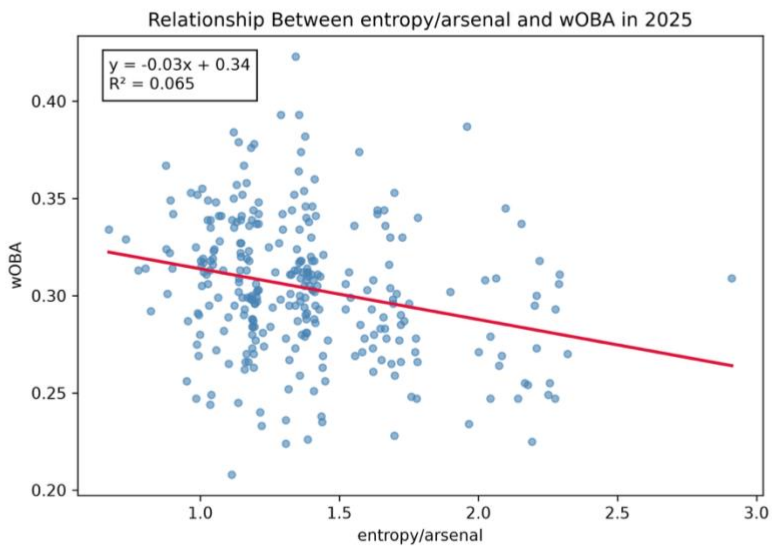
Entropy As A Function Of Arsenal Depth

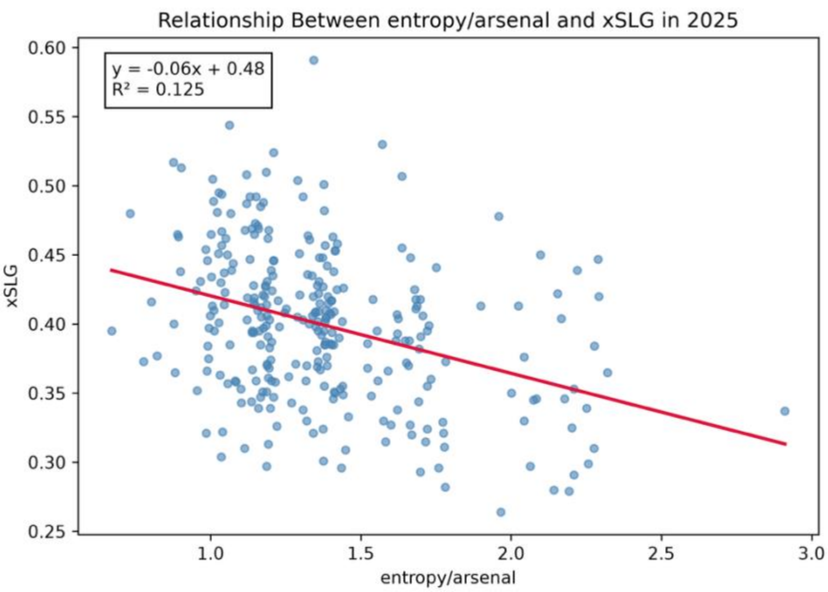


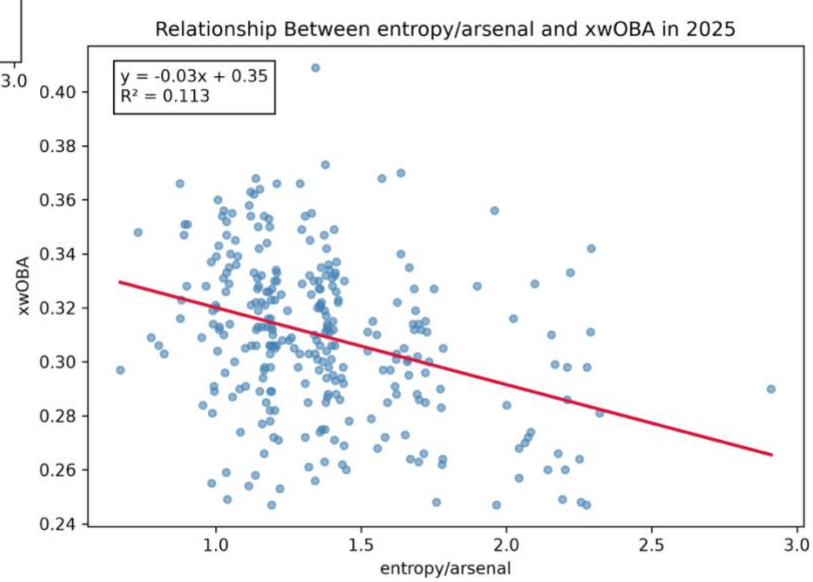





Relationship Between entropy/arsenal and K% in 2025
y = 5.21x + 15.94
R² = 0.121
K%
entropy/arsenal


Relationship Between entropy/arsenal and wOBA in 2025
y = -0.03x + 0.34
R² = 0.065
wOBA
entropy/arsenal


Relationship Between entropy/arsenal and xSLG in 2025
y = -0.06x + 0.48
R² = 0.125
xSLG
entropy/arsenal


Relationship Between entropy/arsenal and xwOBA in 2025
y = -0.03x + 0.35
R² = 0.113
xwOBA
entropy/arsenal